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Q1) choose the best answer

1. The output of a system $y(t)$ is related to its input $x(t)$ according to the relation $y(t) = x(t)sin(2\pi t)$ this system is

a- Linear and time-invariant

b- Linear and time-variant

c- Nonlinear and time-invariant

d- Nonlinear and time-variant

2. Consider the system represented by $y[n] = 2x[n] + 3$

a. The system is linear and follows the additive property.

b. The system is linear and does not have homogeneity property.

c. The system is nonlinear and does not have homogeneity property.

d. The system does not have the homogeneity property but follows the additive property.

3. Consider a signal $x(t) = 5\cos\left(\frac{2\pi t}{3}\right) + 9\sin(0.5\pi t) + 3\sin\left(\frac{\pi t}{3} - \frac{\pi}{6}\right) - 12$, so that

a. $x(t)$ is periodic with frequency (1/12) Hz.

b. $x(t)$ is periodic with period 1.2s

c. $x(t)$ is not periodic

d. $x(t)$ is periodic with frequency 12 Hz

4. What's the even part of the signal $x(t) = 2 + cos(t)$?

a. $2 + sin(t)$ b. ~~$2cos(t)$~~ $2+cos(t)$ c. $2 - sin(t)$ d. $2\,cos(t)$

5. Which of the following is false for a continuous time signal?

a. Any signal can be expressed as the sum of an odd and an even signals.

b. Sum of two periodic signals is always periodic.

c. Any periodic signal can be expressed as sum of sinusoidal signals.

d. None of the above.

$-15) - r(t-15)$

6. If $f(t)$ has an energy E so the energy of $f(2t)$ and $f(0.5t)$ respectively are

a. E and E b. E and $0.5\,E$ c. $0.5E$ and $2E$ d. $2E$ and $0.5E$

7. The following systems $y(t) = \frac{1}{3}(x[n] + x[n-1] + x[n+2])$ is
a. Dynamic & causal b. static & non-causal c. static & causal $\boxed{\text{d.}}$ dynamic & non-causal

8. The value of $\int_{-\infty}^{\infty} e^{-t}\delta(2t-2)dt$ is
$\boxed{a.}$ $\frac{1}{2e}$ b. $\frac{2}{e}$ c. $\frac{1}{e^2}$ d. $\frac{1}{2e^2}$

9. If $x[n] = \{3,2,1\}$ and $h[n] = \left(\frac{1}{2}\right)^n u[n]$ if $y[n] = x[n] * h[n]$ so $y[1] =$ $\frac{3}{2}+2$
a. 0 ↑ b. 3 $\cancel{\boxed{c.}}$ 3/2 d. none of above هيك بالمفروض

10. If $g(t) = \begin{cases} \frac{t+4}{4}, & -4 \le t < 0 \\ \frac{2-t}{2}, & 0 \le t < 2 \end{cases}$, so the $g(0.25t)$ will define in negative time period is
a. $-1 \le t < 0$ b. $-4 \le t < 0$ $\boxed{c.}$ $-16 \le t < 0$ d. all are correct

11. for $\int_{-5}^{6} x(t)\delta'(t-3)dt$ the result will be $= x(t) = t^2 + 2$
a. 0 b. -8 $\boxed{\text{c. 11}}$ d. 1

12. The $sgn(t) =$
a. $u(t) - u(t)$ b. $u(t) + u(-t)$ $\boxed{c.}$ $u(t) - u(-t)$ d. $sgn(-t)$

Answer the questions (13-15) through the convolution between $x(t) = 2$ only for $0 \le t < 1$ and $h(t) = 1$ only for $0 \le t < 2$ where $y(t) = x(t) * h(t)$

13. $y(2) =$ a. 0 $\boxed{b.}$ 2 c. 1 d. 0.5

14. $y(t) = x(0.5) * h(0.5)$ equals to a. 0 $\cancel{\boxed{b.}}$ 2 c. 1 d. 0.5

15. The no overlapping area will be
a. $t < 0$ only b. $t > 2$ only c. $0 < t$ & $t > 2$ $\boxed{d.}$ $0 > t$ & $t > 3$

Q2: Prove mathematically (in detail) that the following signal is $x(t) = e^{j[(\frac{\pi}{2})t-1]}$ (8 Marks)

1. Periodic or $\boxed{\text{aperiodic}}$ signal?
2. $\boxed{\text{Energy}}$ or power signal?
3. Even or odd signal? neither
4. Causal or $\boxed{\text{non-causal}}$ signal?

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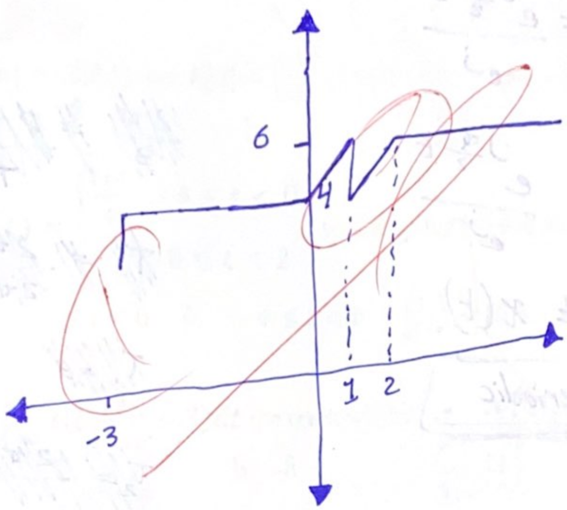
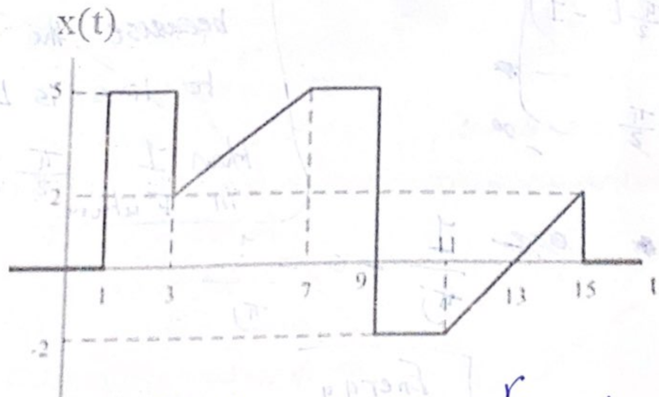
Q3:

I. Sketch $f(t)$ as detailed in the following equation: (4 Marks)

$$f(t)=u(t+3)+2r(t)-u(t-1)-2r(t-2)+3$$



II. write down the signal in mathematical form (3 Marks)



$x(t) = 5u(t-1) - 3u(t-3) + \frac{3}{4}r(t-3) - \frac{3}{4}t(t-7) - 7u(t-9) + r(t-11) - 2u$

$\frac{3}{4}r(t-3)$ $\frac{-3}{4}r(t-7)$

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